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List of Acronyms

FEM Finite Elements Method.	PEM Poroelastic material.
JCA Johnson-Champoux-Allard.	SCM Spectral Collocation Method.
MST Multiple Scattering Theory.	SRR Split-Ring Resonator.

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Introduction

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severely affected areas. Sources of noise can be diverse; in urban environments, concerns are mostly on traffic and population noise. From industrial origins, noise can have various consequences, causing not only risks to the workers and the neighbouring populations, but also structural hazard to installations. Generally, motorized transport is a major issue in all environments, including aircraft, public transportation, roads, commercial vessel traffic, etc.

The European Environment Agency estimated in 2021 that 1 in 5 European resident are exposed to unhealthy levels of noise, its main cause being traffic. These noise exposures can have a strong impact on health¹ favoring impaired circadian rhythm, oxidative stress and cardiovascular dysfunction², as well as psychological consequences like anxiety and depression³. Moreover, anthropic noise pollution also affects biodiversity at large, causing stress, behaviour changes, and masking sensory perception⁴. Marine traffic was shown to generate stress on marine mammals and modify their behaviour by provoking area avoidance and vocalization disruption⁵.

Although sound intensity can be measured objectively, the impact of noise on individuals is in part shaped by social and cultural context. Studies show that expected behaviours norms and customs strongly influence reported annoyance⁶, with comparative surveys revealing clear cross-cultural differences. More recent work has linked collectivistic cultural norms to systematically higher everyday noise exposure as an example^{7,8}. Through these examples, we can conclude that noise is a very broad topic that raises complex questions in various scientific disciplines, regarding its impact on human life and biodiversity at large. In this work, we want to focus on noise mitigation using engineered structures in acoustics.

Most noise sources have emerged from the modernisation of societies through various innovations; with them come with the need to mitigate their noise impact up to our current capabilities, which is able to evolve thanks to research on noise barriers and acoustic insulation systems. Various strategies have been applied throughout the ages. Dating back to 4th century BC, acoustic jars were used to improve the room acoustics in Greek and Roman theatres as well as in the Middle Ages in Western Europe in churches⁹. These resonators are thought to be capable of reducing reverberation time in buildings, covering a frequency range matching that of human spoken voice. As another example, a famous roman amphitheater was shown to have a seat row disposition able to improve intelligibility of speeches from the stage¹⁰.

Contemporary to these antique devices, porous and fibrous materials have been traditionally used to absorb sound in the last century¹¹⁻¹⁶. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature¹⁷⁻²¹. They can be found in natural occurrences or engineered and cover very large use cases of acoustic insulation such as industry, transport, aerospace, concert halls and many others.

State of art

Overview of Acoustic Metamaterials

In more recent years, the emergence of more complex engineered structures has disrupted the landscape of acoustic absorbers. Metamaterials are composite structures designed to exhibit physical effective properties that go beyond the capability of the sole material that composes the bulk. They can be created by a lot of various means, including engineered microstructure, a gradual evolution of the material properties, or repeating patterns or scatterers implemented in one or several directions in space and embedded in a host material as shown for various dimensions in Fig. I.1. Compared to homogeneous layers, they generally lead to more complex structures, and are able to achieve enhanced effects. Usually allowing for a tunable response, these structures offer great flexibility and open new perspectives in material design. This field has been initiated by Veselago, who has theoretically described the electrodynamics of materials having a negative refraction index (occurring in the case of a doubly-negative permittivity and permeability), now known as the Veselago lens in electromagnetic metamaterials²². These discoveries have subsequently sparked interest in various disciplines, such as photonics, optics²³ and in the case that concerns us, acoustics. Metamaterials can be used to design antennas and absorber for electromagnetic waves and seismic protection devices among others. In acoustic, they allow for various applications such as absorption, filtering, diffusion or cloaking.

Seminal works have been done on the study of the use of lattice arrangements or crystals for wave control applications. Photonic crystals are used to achieve the control of light and electromagnetic waves²⁴. Similarly in acoustics, phononic crystals have been developed for the control of elastic and acoustic waves as the 2D (or 3D) periodic distribution of scatterers in a solid host medium²⁵ as depicted on Fig. I.1. In the case where the host medium is constituted by a fluid, these structures are called sonic crystals²⁶. These structures are often studied through their band structure (also called dispersion diagram/relation when applied to the study of non-periodic geometries), which will be discussed in extensive details in Section [ref this in chap1](#). In the remaining of the literature overview, focus will be centered around metamaterials that involve periodicity in only one direction in most cases. These acoustic metamaterials can be divided into two main categories: the first concerns the so-called *reflection problem* where the case study is that of an incoming wave on some sample or panel which has a rigid-backing at the end. There, the consideration is focused on mitigating the reflected wave and achieving important acoustic absorption. The other case is that of a *transmission problem* where the incoming wave is able to transmit through the sample. In this case and depending on the application, either or both reflection and transmission are sought to be mitigated. In each of these cases respectively, the structures can be called metasurfaces and interfaces.

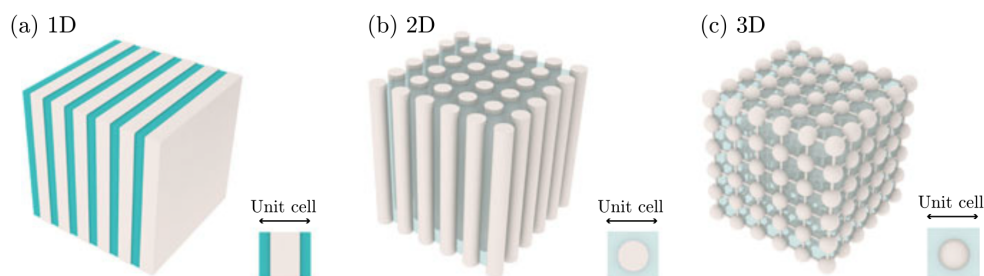


Fig. I.1: Periodic cells in 1, 2 and 3D. Image taken from Ref.²⁷

Structured materials that tackle the reflection problem can involve many various strategies. The combination of a perforated plate and coiled-up channels is able to achieve large broadband absorption underwater²⁸. Micro-perforated plate can also be used to achieve interesting absorption

properties for airborne acoustics²⁹. Materials structured from resonant building blocks with resonance frequencies in the sub-wavelength regime i.e., for frequencies which wavelength is much bigger than the typical size of the material, have also been studied^{30–32}. Large width elastic bandgaps have been achieved using locally resonant cells in the design of a sonic crystal³³. These resonances introduce strong dispersion of the waves and the properties of the system are thus dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions^{34,35}. Balancing the dissipation and leakage of each resonance yields the recently described *critical coupling*. In this case, the reflection drops to zero³⁶. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves^{34,37} and broad range of frequencies^{37–41}. This can be achieved by tuning the structure using the scattering matrix zeros plotted in the complex frequency plane; critical coupling would be achieved when the scattering matrix zero has a vanishing imaginary frequency. Similarly to these broadband absorbers, Yang *et al.* have realized the backing of a porous layer using Fabry-Pérot channels to design an efficient structure⁴².

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been widely developed in the literature, a small subset of these works include Refs. ^{43–46}. These configurations are based on resonant elements and display a zero transmission in the lossless case at their resonance frequency. In reality, losses cannot be avoided and a small amount of the energy is always transmitted. Membrane-type acoustic metamaterials^{47–49}, or acoustic metamaterials made by Helmholtz resonators^{50–52} have been used to develop systems with high values of transmission loss for audible acoustics. Considering the vibroacoustic coupling allows for more precise depiction of real-life applications. Works on periodic soft inclusion in a soft solid matrix was done by Aberkane *et al.* showing improved transmission loss affected by the presence of soft inclusions in the solid layer. In this case, the presence of losses was able to improve the transmission loss at low frequencies. Also accounting for vibroacoustic coupling, attached resonators to walls have been also used to reduce the transmission loss of panels^{53,54}.

For the case of mirror-symmetric structures, it is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, then only 50% of energy can be absorbed as shown in several articles^{55,56}. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances^{47,50,57}, although requiring for a compromise between absorption and thickness. A more complex approach consists in designing degenerate resonators. By using combinations of membrane and cavity resonators^{47,58} and with slits loaded with Helmholtz resonators⁵⁶, quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides⁵⁵. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side^{34,37,49}. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber³⁸.

From porous layers to metaporous surfaces

As mentioned in the beginning of the introduction, porous materials have been traditionally widely used for their acoustic insulation performances. These materials are lossy: an airborne acoustic wave propagating through a porous layer is dissipated as it propagates in the bulk in the form of

viscous friction due to the drag between the air and the pore walls and in the form of heat due to the small ducts and the existence of a thermal boundary layer. These losses mechanisms are strongly dependant on the nature of the material, as they are mainly affect by the geometry of the microstructure. Common models usually account for homogenized effective parameters to characterize these materials as will be discussed in a further section. Since wave velocity in porous media tends to be smaller than that of air, allowing to lower slightly the frequency of the quarter-wavelength resonance, which corresponds due the frequency where such materials usually start to have an efficient acoustic absorption. Section 1.1 is dedicated to presenting the various models used for porous and poroelastic media modelling. Their absorption efficiency depends thus on their inherent losses but also on the impedance mismatch between the layer and the medium from which the impinging acoustic wave is originated (i.e usually air). The acoustic impedance mismatch leads to reflection at the interface, while the energy transmitted through the interface in the porous layer gets attenuated. In real-life applications, these materials are not closed systems and there is a need to account for their coupling irl to surrounding domains into which part of the acoustic energy is susceptible to be radiate back into the half-space.

In order to increase the capabilities of acoustic absorption of porous materials, some works can involve the modification of the core structure of porous and poroelastic materials by tuning of the microstructure, introducing a gradient of the properties, or by using anisotropic materials. Recently, with the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties^{59,60}, or on anisotropic materials^{61–64}. Recently, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption⁶⁵. Other configurations have been studied, among them the periodic partitioning of the metaporous layer⁶⁶ or a labyrinthine channel embedded in a porous host⁶⁷, based on the idea of coiling up the space of the channel as was done for non-porous metamaterial designs. Recently, a hierarchical multilayer metaporous structure was proposed using an optimization procedure, giving rise to gradient and periodic designs⁶⁸.

Some studies involving porous layers embedding periodic arrays of inclusions were done during the last 15 years^{69–72}. These inclusions can have either resonances or not and they allow for an increased dissipation in the viscous regime. These structures are refered to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems^{73,74}. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness⁷⁰. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs⁷⁵. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary^{71,75}. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled⁶³, designed and experimentally characterized showing perfect absorption peaks. More complex spiral-shaped slits have been embedded in a porous layer and have shown to be able to maintain a high absorption for large incidence angles⁷⁶. Xiong *et al.* have studied a metaporous liner with embedded resonator and shown the existence of an exceptional point when the geometry is tuned properly, leading to enhanced absorption⁷⁷.

Fig. I.2: Example of metaporous configurations

Previous metaporous layers studies have considered a host medium modelled as a rigid-frame porous layer; extensions of these structures to consider poroelastic materials have been done⁷⁸. Interesting properties were exhibited for thin elastic shell inclusions, and particularly in the case of a coupling between the trapped and the first volume mode leading to a high wide band absorption. While this previous study implements a semi-analytical model that makes use of Multiple Scattering

Theory (MST), it is worth noting here that different modelings exist as well, based on a Finite Elements Method (FEM) and MST have been validated for the case of an air-saturated metaporoelastic configuration with a solid inclusion and rubber coatings⁷⁹. This thesis is especially focused on the computation of the dispersion relation of such configurations. Therefore, a dedicated literature review dealing with the tools available to achieve such computation will be given in Chapter 1.

Wave propagation modelling and dispersion analysis

This section should introduce scattering problems vs dispersion problems

The analysis of wave propagation in periodic and complex media often relies on the computation tools. Two main topics are of interest in this thesis: the scattering of an incident wave by a structure as well as its dispersion relation. The first one is simply the acoustic response of a system while dispersion relations are diagrams that represent the wavenumber in a given direction against the frequency, thus informing on the evolution of the wave velocities in the structure and thus the dispersion within. In the context of metamaterials and periodic structures, they are often referred to as band structure.

A theoretical definition of a dispersion relation consists in the spectrum of all solutions of a set of equations given a set of parameters. It shows the possible existing states of a system. Generally the form of the dispersion relation, and the various modes that are uncovered can give very in-depth details about the modal behaviour of any structures. In ultrasonics and non-destructive testing, localizing zero-group velocity (ZGV) points in the dispersion relation is very helpful for a number of uses including defects localization and material characterization⁸⁰.

Scarce literature focuses on the computation of the dispersion relations of dissipative layers in general exists: the dispersion of waves in a single poroelastic layer has been studied by Boeckx *et al.*^{81,82} while an unpublished paper from Kiefer *et al.* took interest in Lamb waves in viscoelastic layer coupled to surrounding half-spaces⁸³. There is an interest in analogously studying the coupling between surrounding media and metaporous surfaces, as part of the acoustic energy of the wave propagating in the surface should radiate back into the domain, leading to more complex loss effects on the structure.

These aspects will be discussed in detail in the following chapter, both from the theoretical perspective of porous and poroelastic media (and their coupling to surrounding media) and in terms of the available numerical tools for their computation.

Motivations & Work Objectives

This thesis is dedicated to the numerical modeling of wave propagation in open-system structures comprised of stacks of layers, homogeneous or with embedded scatterers. As it has been seen in the previous section, these materials combine the dissipative characteristics of porous media with the geometric or structural features typical of acoustic metamaterials, thus offering advanced control over their scattering properties. Understanding and predicting the acoustic behavior of such systems is therefore essential for designing efficient structures. In particular, this work focuses in part on the calculation of the dispersion of waves in these structures, and understand how the propagating and evanescent modes that exist in these configurations are shaping the scattering properties. Thence, the development of efficient numerical strategies is critical to this study.

The tuning of metamaterials could be achieved in some studies by looking at the zeros of the reflection coefficient in the complex frequency plane, thus highlighting the zeros of the scattering matrix. However, the wavenumber might be able to give out additional informations regarding the acoustic behaviour of metaporous layers. Band structure gives useful informations concerning sonic and phononic crystals. However, the dispersion relation of a metaporous layer embedded in an open system has never been addressed; both mechanisms of energy leakage in the surrounding media and

dissipations in the porous layer should be investigated. However, this imply a heavy modification of the boundary conditions of the structure. It expected to have significant consequences on the propagation of modes therein. This work will be done on both homogeneous layers as well as periodic structures.

From these observations, we should review existing numerical methods able to compute the acoustic response of metaporous structures and consider ways to obtain a precise and robust method able to predict the scattering properties and acoustic responses of metaporous surface and interfaces. Additionnaly, we should evaluate methods that have been implemented for the computation of guided waves in layers and stacks of layers coupled to semi-infinite media. Confronting the dispersion relation, with the absorption coefficient should gelp highlighting the modal behaviour responsible for the various absorption peaks obtained by previously studied configurations and also highlight the full modes existing in the layers which might uncover additional phenomena.

Contents of the thesis

This introductory section has set the context as well as an overview of the literature in which this work is situated. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient that the traditional foams used for sound insulation. In particular, metaporous surfaces have shown to be promising options achieving one or multiple unitary absorption peaks at subwavelength frequency. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of these configurations. Thus, in this thesis the main interest and final is to succesfully compute the dispersion relations of guided waves in such structures. The manuscript is split into four different chapters as follows,

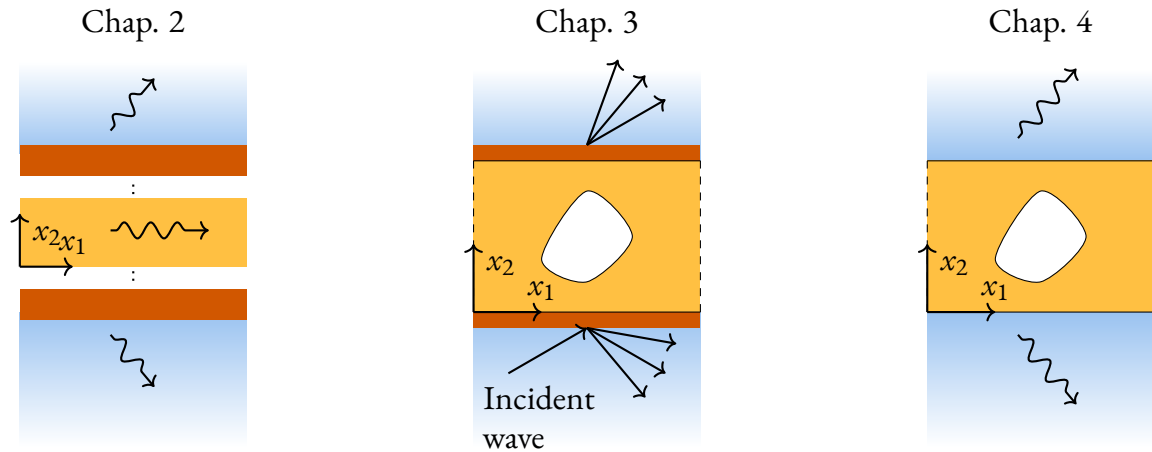


Fig. I.3: Diagram presenting a general layout for the configurations that will be studied in each chapter

- CHAPTER 1 introduces the fundamental concepts necessary for the good understanding of the thesis. This chapter focuses on the scattering of waves on dissipative layers and the guided waves formalism that is used for the calculation of dispersion relations. Additional bibliography is proposed on these topics as well as a summary of usable numerical methods to compute such problems.
- CHAPTER 2 targets the study of the propagation of guided waves in multilayer arbitrary structures with dissipative layers. A simple numerical method, the Spectral Collocation Method (SCM) is implemented and extra steps allow for the local coupling of surrounding media allows for the computation of the dispersion relation for these configurations. This chapter is constituted from paper⁰.

- CHAPTER 3 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A global method is coupled to a FEM. Results focused on the performance of the proposed method, compared to other similar frameworks is proposed. The convergence of this method is shown for various metaporoelastic surfaces, which geometries were studied in previous papers. This chapter is originally an article referenced as paper. [cite when submitted](#).
- In CHAPTER 4, the modal behaviour of metaporous structure is studied. The fem method from the previous chapter is adapted to describe this problem. Results are presented for the case of a rigid inclusion, as well as with embedded Split-Ring Resonator (SRR). Contents for this chapter were published in paper. [cite when submitted](#).

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This theoretical chapter lays the foundation for the thesis, giving some notions of the fundamental necessary aspects, in order to guide the reader through the various chapters that will follow. At first, an overview of the modelling in porous media is proposed.

1.1 | Modelling of porous media

Porous materials are constituted of two distinct domains: an elastic matrix Ω_s with a pore network Ω_f , which is filled by some saturating fluid, they are biphasic. A diagram of their structure is shown in Fig. 1.1. The porosity is defined as the fraction of fluid over the total volume of the material, as $\phi = V(\Omega_f)/V(\Omega)$. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. The geometry of the fluid phase forms a complex microstructure at a microscopic level. Fortunately, at acoustic scales, these materials can often be considered at a meso-/macroscopic scale and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that account for their complex inner geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through friction (viscous losses), thermal exchanges and elastic dissipation when the skeleton and the fluid are coupled. These losses are thus dependant on the geometry, that is the porosity, pore network size, connectivity and shape, and the frame mechanical parameters.

1.1.1. Rigid-frame approximation

In the so-called rigid-frame case, the skeleton of the porous material is motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{i\omega t}$ throughout this manuscript (with the exception of

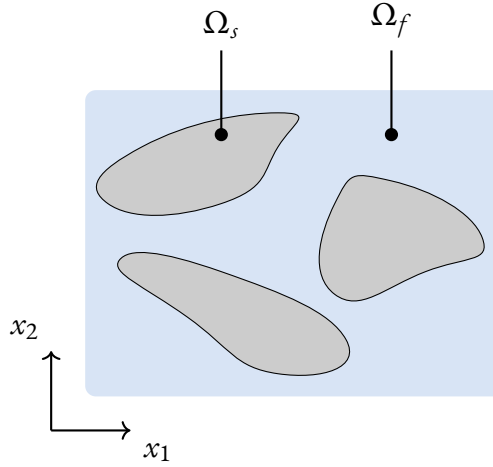


Fig. 1.1: Porous material structure

Chapter ??) The Helmholtz equation governs the propagation,

$$(\nabla^2 + k_{eq}^2) p(\mathbf{x}) = 0, \quad (1.1)$$

where p is the pressure field and k_{eq} is the wavenumber of the equivalent fluid. In the case where porous or fibrous materials have a porosity close to 1 and the frequency range of interest is limited, the Delany-Bazley model was originally developed in 1970 to propose empirical expressions for the values of the complex wavenumber and characteristic impedances^{11,84} and later improved by Miki⁸⁵. Although limited in accuracy and validity domain, these models have been widely used since they only required the flow resistivity of the porous material. Later on, a more precise semi-phenomenological model additionally accounting for the viscous and inertial dissipative effects was developed from 1987 to 1991. It is the Johnson-Champoux-Allard (JCA) model which models the complex effective density $\rho_{eq}(\omega)$ accounting for the frequency dependant viscous losses and effective bulk modulus of the material K_{eq} accounting for thermal exchanges in the layer as,

$$\begin{aligned} \rho_{eq}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - i \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right), \\ K_{eq} &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) / \left(1 + \frac{i R_f \phi}{\rho_f \alpha_\infty \text{Pr} \omega} \sqrt{1 - i \text{Pr} \omega \rho_f \mu_f \left(\frac{2 \alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}. \end{aligned} \quad (1.2)$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f , kinematic viscosity μ_f , pressure P_0 and Prandtl number $\text{Pr} = c_p \mu_f / k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f \phi / (\rho_f \alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2. Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is

coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory^{86–89} is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works¹¹.

Before presenting the full Biot model, some quantities should be provided additionnaly to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1 - \phi)\rho_s$. The bulk moduli are denoted K_b and K_s , respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\tilde{N} = N(1 + i\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is thus $\rho_2 = \phi\rho_f$ and the inertial interaction density defined by $\rho_{12} = -\phi\rho_f(\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. ??) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation^{86,88,89} depend on the solid displacement vector $\mathbf{u}_s$ and the fluid displacement $\mathbf{u}_f$,

$$\begin{aligned}\nabla \cdot \boldsymbol{\sigma}_s(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{11} \mathbf{u}_s - \omega^2 \tilde{\rho}_{12} \mathbf{u}_f, \\ \nabla \cdot \boldsymbol{\sigma}_f(\mathbf{u}_s, \mathbf{u}_f) &= -\omega^2 \tilde{\rho}_{12} \mathbf{u}_s - \omega^2 \tilde{\rho}_{22} \mathbf{u}_f,\end{aligned}\tag{1.3}$$

where $\boldsymbol{\sigma}^s$ is the solid stress tensor, $\boldsymbol{\sigma}^f$ the fluid stress tensor and the apparent densities of the solid-phase is $\tilde{\rho}_{11}$, that of the fluid is $\tilde{\rho}_{22}$ and the coupling apparent density is $\tilde{\rho}_{12}$. They are complex quantities expression that also account for the contribution of viscous forces and are given as,

$$\tilde{\rho}_{22} = \phi^2 \tilde{\rho}_{eq}, \quad \tilde{\rho}_{12} = \rho_2 - \tilde{\rho}_{22}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}.\tag{1.4}$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned}\boldsymbol{\sigma}_{s,ij} &= [(\tilde{P} - 2\tilde{N})\nabla \cdot \mathbf{u}_s + \tilde{Q}\nabla \cdot \mathbf{u}_f] \delta_{ij} + 2\tilde{N}\boldsymbol{\epsilon}_{s,ij}, \\ \boldsymbol{\sigma}_{f,ij} &= (\tilde{Q}\nabla \cdot \mathbf{u}_s + \tilde{R}\nabla \cdot \mathbf{u}_f) \delta_{ij} = -\phi p,\end{aligned}\tag{1.5}$$

where $\nabla \cdot \mathbf{u}_s$ and $\nabla \cdot \mathbf{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and $\boldsymbol{\epsilon}_s$, is the solid-phase strain tensor. The Biot coefficients are denoted as $\tilde{P}$ for the solid elastic coefficient, $\tilde{Q}$ for the elastic-fluid coupling coefficient, and $\tilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\tilde{P} = K_b + \frac{4}{3}N + (1 - \phi)^2 \frac{\tilde{K}_{eq}}{\phi}, \quad \tilde{Q} = \tilde{K}_{eq}(1 - \phi), \quad \tilde{R} = \phi \tilde{K}_{eq}.\tag{1.6}$$

The full coefficients are generalized forms required when the limit $K_b/K_s \rightarrow 0$ no longer holds, as is the case for water-saturated PEM.

A different formulation, especially convenient for finite-elements discretization is known as the $(\mathbf{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\hat{A} = \tilde{P} - 2\tilde{N}_s - \frac{\tilde{Q}^2}{\tilde{R}}, \quad \hat{\rho} = \tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}^2}{\tilde{\rho}_{22}}, \quad \beta = \phi \left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}} \right).\tag{1.7}$$

β is another coupling coefficient, also denoted as $\tilde{\gamma}$ in the literature. Using the second row of Eq. (1.3) allows for the expression of the fluid displacement as, $\mathbf{u}^f = \phi / (\tilde{\rho}_{22}\omega^2) \nabla p - \tilde{\rho}_{12} / \tilde{\rho}_{22} \mathbf{u}^s$ and using that in Eq. (1.3) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\mathbf{u}_s, p)$ formulation of the Biot theory⁹⁰ is expressed as,

$$\begin{aligned} \nabla \cdot \hat{\sigma}_s + \hat{\rho}\omega^2 \mathbf{u}_s + \beta \nabla p &= 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\beta}{\phi^2} \nabla \cdot \mathbf{u}_s + \frac{1}{R} p &= 0, \end{aligned} \quad (1.8)$$

where $\hat{\sigma}_s$ is the *in vacuo* solid stress tensor,

$$\hat{\sigma}_{s,ij} = \sigma_{s,ij} - \phi \frac{\tilde{Q}}{R} p \delta_{ij} = \hat{A} \nabla \cdot \mathbf{u}_s \delta_{ij} + 2N_s \epsilon_s \quad (1.9)$$

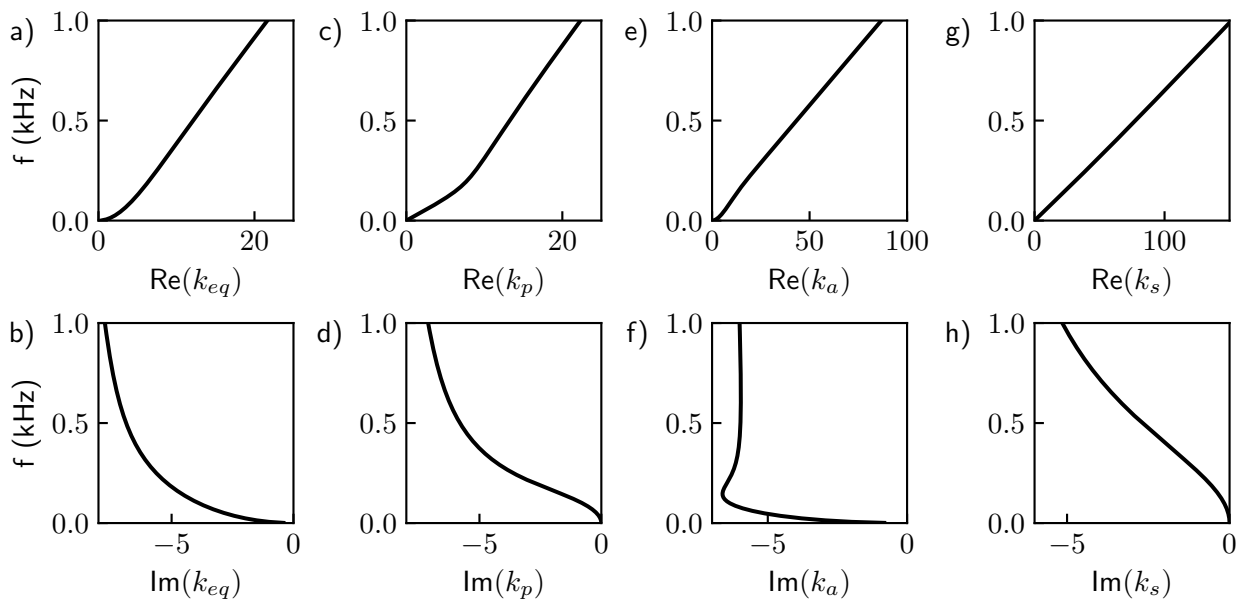


Fig. 1.2: Wavenumbers in unbounded porous media. a) Equivalent fluid wavenumber. Wavenumbers and attenuation for each of the 3 Biot waves.

These various formulations provide several way of modelling equivalently wave propagation in poroelastic media. It appears that PEM supports three waves: a compression and shear wave related to the solid phase as well as an additional compression wave related to the fluid phase. ++ **détails** Following the section 6.5 of the book of Allard and Atalla¹¹, the three wavenumbers of these waves are (denoted as $\delta_i, i = 1, 2, 3$ in the text),

$$\begin{aligned} k_{p,a} &= \sqrt{\frac{1}{2} \left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t \right)} \\ k_s &= \omega \sqrt{\frac{\hat{\rho}}{N}}, \end{aligned} \quad (1.10)$$

with,

$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}}, \quad \delta_1 = \omega \sqrt{\frac{\hat{\rho}}{\tilde{P}}}, \quad \delta_2 = \omega \sqrt{\frac{\hat{\rho}_s}{\tilde{P}}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2 \delta_1^2}. \quad (1.11)$$

The propagation characteristics, i.e. wavenumbers of the waves propagating in porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber

for the equivalent fluid as well as the three for the poroelastic layer are complex. Their real part accounts for the propagation of the phase flow, inversely proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials.

1.2 | Methods for modelling metaporous structures

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of wave propagation in unbounded porous and poroelastic have been studied and shown that these materials support for the propagation of respectively one and three distinct waves. The present section proposes the introduction of a geometry in the system, that is to say boundaries or interface conditions delimiting different domains characterized each by different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, which impinges at the surface of a structure and is scattered (reflected and transmitted) through each interface.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is referred to as a layer. The waves propagate by reflecting alternately from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often pose problems that are difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries.

1.2.1. Wave propagation in homogeneous layers

Introducing a simple problem, that of the scattering of an incident plane by a porous layer. The incident pressure field takes the form of,

$$p_i = ae^{-ik_1x_1}e^{-ik_2x_2}, \quad (1.12)$$

with wavenumbers such that they hold for the Snell-Descartes law,

$$k_1^2 + k_2^2 = k_0^2, \quad (1.13)$$

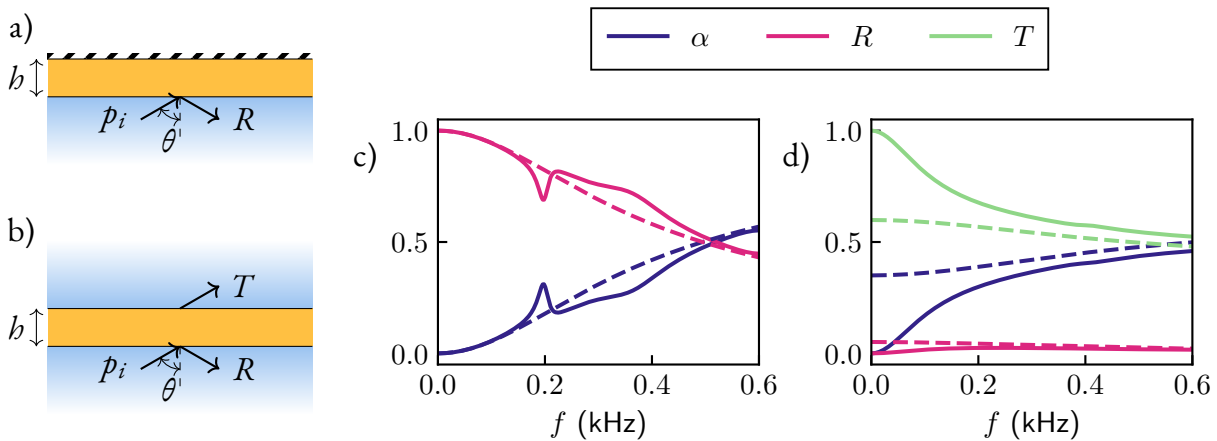


Fig. 1.3: Diagram for the a) reflection, b) transmission problem. Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) for the c) reflection, b) transmission problem.

Global Method: plane wave modelling

The analytical description of the system amounts finally to a linear system built from the interface conditions, which unknown vector are the amplitudes of each of the existing waves in the system. The forcing right-hand side corresponds to the incident plane. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in a symmetric configurations are represented by the diagrams in Figs 1.3a and b and results in Figs 1.3c and d, respectively. There, the layer for both porous and PEM are both $b = 5$ cm thick and the incident wave propagates with a $\theta = 30^\circ$ angle. The parameters for each material corresponds to the one used in Table ?? from Chapter ?. The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but

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